Unit 3 - Impact of IT & Information Systems
Assignment 5

Introduction

Scenario / Vocational Context - *Now that iTech Consultancy has become established and is building a diverse client portfolio, it is often required to offer bespoke solutions to clients that enhance and improve their organisations through IT enabled business systems and Managed Information Systems (MIS). As such your Manager would like you to produce a comprehensive proposal for IT enabled improvements for prospective clients.*

Before writing your proposal you must first create your own case study; through prior in class activities and liaising with your Tutor.

Using a scenario agreed with your tutor, you must complete a comprehensive ICT proposal containing potential ICT improvements. Your proposal should be completed in MS Word and be in a form of a professional report, with a front page, contents page, titles and subtitles throughout.

Task 1

(U4:P6) - Produce a proposal for an IT-enabled improvement to a business system.

At least two new pieces of hardware, that improve existing systems

With an organisation like the Jersey Aero Club, it obviously wouldn't sell any products to the customers. Instead, the business focuses on teaching beginner pilot how to improve their aviation skills.

So, with the management of many inexperienced and experienced people, technology can prove to be very beneficial to an organisation with standards like this. Technology which ranges from software and hardware can be very useful to this organisation, but it depends on what type of software and hardware is to be implemented due to some being more useful than others.

One type of new hardware that would improve existing systems within the Jersey Aero Club would be an implementation of a CPS, which is a classroom performance system. Since Jersey Aero Club teaches trainee pilots how to fly, both in a classroom and an aircraft, in which the implementation of a CPS would be beneficial within the classroom of the trainees.

The CPS can be used daily to monitor understanding and progress on goals which the trainees are responsible for achieving in that specific deadline or time. This could be seen as an essential for class use when teaching the trainee pilots the essentials and keeping a record on how each of them are performing.

Another type of new hardware that would improve existing systems within the Jersey Aero Club is getting a tablet or any other form of mobile device. This piece of hardware can be used for many different jobs, due to the wide selection of applications that can be downloaded within the mobile devices.

The mobile device can be used to keep schedules, dates, reminders and store many forms of data such as images, videos and other information which can be useful for the organisation. This could be seen as an essential for outside class use when teaching the trainee pilots aviation and also keeping a record on how each of them are performing.

At least two new types of software or social media

One type of new software that would be an improvement to the organisation is a database software which can be used to store plenty of useful information. Each database software is made and used differently, so it means that some are better than others for containing information, but they all do roughly the same job.

This software can be used, with the information gathered, to determine which trainee has good reports, the flight progression of each trainee, which trainee is ready to receive a flying license and even which aircraft is to be purchased, delivered and used.

Another type of new improvement to the organisation is the use of social media, such as facebook or twitter. Since twitter and facebook are both social medias which are widely used in the modern day, it would benefit the organisation if they used these forms of social media to their benefit.

The social media can be used to promote the organisation and to alert members of the JAC for any upcoming events. The reports of upcoming events is also a benefit because not only could it bring current members of the JAC to attend, but could also bring newer members to the JAC. This type of promotion can bring in new members who are interested to the JAC, meaning that they could benefit from the income from the new members.

Additional software and hardware required to utilise your above suggestions

Even though the hardware and software which can be proven to be beneficial to the company is important to have, they wouldn’t be beneficial without the hardware and software which is required to utilise them.

For the CPS, the staff in control of the CPS wouldn't really need an additional software or hardware in order to utilise it, since the CPS is a package in which it is already good to use when brand new. However, the organisation would need a database and a computer in order to store the information gathered through the CPS in order for it to be useful.

For the mobile devices, it would need different types of software which can be used for the benefit of training and keeping track of trainee pilots. Since mobile devices have become a significant product in the modern day, it wouldn't come as a surprise to see that there are applications in which it can store data of each member and employee of the organisation, which can be stored within the cloud.

For the database software, the manager in control of the database would obviously need a computer system in order to access and use the database to its intended purpose. Depending on the type of database program and the size of the information stored, it could change how high end the PC would have to be in order to have optimal performance when handling the data and the software.

For the social media, the staff in control of the social media would also need a computer system in order to access and use the social media to its intended purpose. Since social media is an internet based software and it is as demanding as the browser used to utilise the social media, means a high end PC would be very unnecessary to use for this type of job.

A basic estimation of cost for each improvement, both initial and ongoing

With the hardware, software and the other additional systems that are required to utilise the suggestions, it would definitely not come in cheap for the organisation. Since each piece of hardware and software vary in price massively, even depending on the brand that produced the product, it would be difficult to get an accurate price for each improvement, which is why a basic estimation is needed.

For the CPS, the basic estimation of £50 for 31 CPS remote clicker pads, meaning that a single one would cost an estimation of £2. This cost wouldn't be very impactful for an organisation, but it also depends on how many members are attending the club to take flying lessons. So, the more members there are training to be a pilot, the more this will cost the organisation.

For the mobile devices, a single tablet such as an iPad Mini 4, would have a cost of around £300 to £400 depending on the retailer. This can be a bit costly to the organisation, especially when a large amount of iPads were to be purchased and used by the staff to use as monitoring and recording data. So, the more staff there is in the organisation, the more this will cost for the organisation.

For the database software, an advanced database software made for a business or organisation, such as "naturalhr" would have a yearly cost of £1,800 for the basic version of the database software. This seems to be very costly for the organisation, but it's a yearly subscription. Meaning that if the organisation can make a lot more than £1,800 a year, then it wouldn't have any significant problems paying for the specific yearly subscription.

For the social media, there isn't really a cost of having to utilise facebook or twitter, since they are free and open to the public. This means that the company would not have to worry on spending money for this area, but there is payment needed to access the social media. Such as purchasing PCs, maintaining the PC and paying for the monthly broadband to access the social media sites. So, the costs of social media is free, but the costs of utilising it goes on top such as PCs, which can range from £200 to £500, and broadband, would could cost around £30 per month.

Effects on staff

There are many different effects that these implementations can have on staff and members, due to everyone being different, some being beneficial. However, there are some effects that it will have on the overall staff due to common problems or problem due to human limitations, but these effects can be overcome and even outweighed by the positives.

One effect on the staff, caused by the CPS, is that the staff can monitor the trainees a lot easier than previously before, proving to be a lot easier to pick which trainee is best for the aviation training.

Which means that the training process can be easier for the staff and much more efficient in picking the best trainees from the worst, so the staff could go and work on the worst trainees to get them in line with the rest of the well performing ones.

However, a negative effect is that the staff members might not be comfortable using them or that they feel they could teach better without them. This effect could be problematic for the organisation and the staff due to some staff members being hesitant on using the product, which can be a problem on gathering information and making a good record on a trainee.

In theory, this would seem to be a rare case because the staff members would use the CPS knowing that it would be recording how well they are doing. Even though this effect is possible, it's only one of the very few effects that it can have on staff members and even trainees.

Another effect on the staff, caused by the mobile devices, like the CPS the staff can monitor the trainees a lot easier than previously before, proving to be a lot easier to pick which trainee is best for the aviation training.

Which again means that the training process can be easier for the staff and much more efficient in picking the best trainees from the worst during aviation, so the staff could go and work on the worst trainees to get them in line with the rest of the well performing ones.

However, a negative effect is that the staff might not be comfortable using them or that they feel they wouldn't really need them, just like the trainees using the CPS. This effect again could be problematic for the organisation and the staff due to some of the staff being hesitant on using

the product, causing the organisation to lose on the investment they made on the product and that the data would not be recorded efficiently.

Again, this would seem to be a rare case because the staff would use the mobile devices knowing that it would be recording how well the trainees are doing and what should be improved. Again, even though this effect is possible, it's only one of the very few effects that it can have on staff members and even trainees.

Another effect on the staff, caused by the database software, is that the staff, such as the database manager, can now easily sort the data on each member of the JAC, the aircraft that the organisation owns and even the staff members themselves.

Which means that the manager in control of the database could now easily and efficiently sort data, which can be used for important information within the organisation to find out about many different problems or benefits that the organisation has, improving on the problems encountered or finding out how they got the specific benefit easier than previously.

However, a negative effect is that the staff might be nervous on what data should be recorded or not know how to record the correct data to be stored within the database. This effect could be problematic for the organisation and the staff due to the possibility of the staff again not recording the correct data, which means that the staff would have a risk of being replaced for another member of staff due to their incompatibility of recording the correct data for the organisation, which can be very problematic for the organisation.

This effect could be overcome with instructions on what data the staff should be collection for the database, which can reduce the worry and concern that might have an effect the staff. Without the worry of the wrong data, the staff could record the data of other staff and trainees with ease.

Finally, the last effect on the staff, caused by the social media, is that the staff can now recruit new members for the JAC and state new events that are coming up a lot easier, instead of the staff having to tell the members physically or giving out brochures to the public to promote the organisation.

Which means that the gathering and communication of members is a lot easier than previously due to the convenience and speed of the social media sites such as twitter and facebook. Which twitter can be used by the staff to report events or to respond to feedback and facebook can be used to also report events, respond to feedback and also promote the JAC for recruitment, without the staff having to go to the public to do so.

However, a negative effect is that the staff might need to recruit new staff members or get staff members which are experienced in operating and maintaining the social media sites. This effect could be problematic for the organisation due to the staff not all being able to use a social media

site, meaning that the company would have to look for newer members of staff or current members of staff to work overtime or to be replaced in order to use the social media sites, which has a benefit of being problematic.

This can have a problematic effect due to the fact that staff in the JAC can be valuable due to their experience with aviation, which is the reason they are working as staff in the JAC and teaching beginner pilots. Meaning that the organisation would have to either get newer staff members, which can increase the budget for the wages of the staff, or that they get current staff to utilise social media, which means that they would have to work overtime or that a different member of staff would have to take over.

Any Knock-on effects to be considered by the business

Even with all the system benefits that the organisation can have, there are some knock-on effects which is possible that the organisation should consider before it could have a significant impact, which can cause the organisation to lose a significant amount of profit.

One of these knock-on effects caused by the CPS, is that the data recorded by the CPS may not be very accurate or might not be transferred properly, either causing the storage of incorrect data or no data at all.

This knock-on effect can prove to have a significant impact because the lack of data for a member of the JAC would mean that progression wouldn't be very possible due to the lack of insufficient data to do so. This means that the member of the JAC might leave the organisation or complain to the staff about the data not being correct, giving a bad reputation for the organisation.

Another of these knock-on effects caused by the mobile devices, is that the data recorded on the mobile devices during the aviation training for the JAC trainee may be incorrect, due to many reasons, or not collected properly, due to technical errors.

This knock-on effect can prove to have a significant impact because, just like the CPS, the lack of data for a member of the JAC would mean that progression wouldn't be very possible due to the lack of insufficient data to do so. This means that the member of the JAC might leave the organisation or complain to the staff about the data not being correct, again giving a bad reputation for the organisation.

Another of these knock-on effects caused by the database software, is that the data recorded in the database may not be very accurate, meaning that the data stored is either incorrect data or the lack of data.

This knock-on effect can prove to have a significant impact because, just like the CPS and mobile devices, the lack of data for a member of the JAC would mean that progression wouldn't

be very possible due to the lack of insufficient data to do so. This means that the member of the JAC might leave the organisation or complain to the staff about the data not being correct, yet again giving a bad reputation for the organisation.

Finally, the last of these knock-on effects caused by the social media, is that it could not be as successful as intended or that any members of the staff aren't willing to utilise and maintain the social media pages for the JAC organisation, which means that the use of social media could either be canceled or very crude.

This knock-on effect can prove to have a significant impact because the absence of a social media means that the organisation won't be able to communicate with the members of JAC or recruit new members very well. Which means that the organisation would stay relatively small and not earn as much profit as they potentially could with the use of social media.

(U4:M3) - Demonstrate originality in proposing an IT-enabled improvement.

Original and new suggestions, using contemporary and forward thinking technologies

Since technology is always changing and having different aspects and characteristics to them, it's possible that contemporary and even forward thinking technologies can have tools in which it proves to be much more efficient.

When compared to the original IT-enabled improvement planned for the organisation, the forward thinking technology could either exceed or prove to be obsolete to have. Meaning that not all forward-thinking technology can be more beneficial than the previous, original technology planned for the organisation.

Such technology can range from better ways of recording the trainees instead of a CPS, innovative mobile devices and even an AI system which can secure the database system and social media information. This may seem plain and simple, but if done correctly can be very efficient and beneficial for the organisation.

For the recording of trainees, instead of using a CPS, the organisation could invest in a high quality surveillance camera for each room instead of them having to hold a remote which provides information.

With higher quality surveillance cameras and integrated microphones, the class of trainees could be recorded and evaluated in much more detail when compared to the CPS, due to the limitations of the CPS having multiple buttons. Where a camera could record anything that the trainee is saying within the class.

This technology can also be used to evaluate each trainee for his/her knowledge better because they have the freedom to answer in any possible way and still be recorded, while a CPS only has limited functions on how to answer questions.

For mobile devices, we have smart phones, tablets and even smart watches, which are common pieces of technology due to their popularity and innovation. However, if the company were to invest on a new forward thinking mobile device, then augmented reality would be a good investment.

AR, or augmented reality is basically the device using a camera looking into the real world, but is using the software to render a virtual object mixed within the real world captured by the camera.

For example, with AR, the organisation could use it to show a model of an aircraft, lecturing the trainees about the vital parts of an aircraft to the new members, but this is just an example. There are many other possible ways in which this could be used to train the new members of the organisation.

However, the use of AR for mobile devices to teach instead of gathering information could have possible problems, but they aren't very likely since there are others means of gathering information.

Using a computer is a function which most people in the modern day can do. However, there are different types of intelligence that can also use a computer system, and that includes AI. When using AI within an organisation, there are many benefits on using one but there are also significant risks as well.

Basically, if AI were used within the organisation, it would be used to keep the database secured and set up in order, and only accessed through voice recognition of the manager. However, it could also be used to write automated messages on the social media account, like so called "bots" which can be found in large businesses.

AI, or artificial intelligence is basically a computer system which is able to perform tasks which would normally require a human to do so. Aspects such as speech recognition, problem solving, translation and even visual perception.

For example, with AI, the organisation could use it to keep the data in the database secure, sending messages within the social media network. However, this is only just an example. There are many other possible ways in which this form of intelligence can be used within a business.

However, the use of AI for a computer system instead of a human operating it could have possible problems, such as the AI malfunctioning, errors within the maintenance of the AI or any other problems in which the AI would cause due to the imperfections.

A longer term proposal, that will involve year on year upgrades overtime

With the noticeably rapid improvement of technology, it's certain that the organisation will have to involve with year on year upgrade cycles. This will cause changes in plans within the organisation, to make overtime proposals to include these year on year upgrade cycles.

These upgrade cycles can prove to be very time and money consuming, especially when the upgrade takes place in a large computer system. Which is why proposals are made to make overtime to carry out these upgrades on the computer systems.

The proposal for the year on year upgrades could vary depending on the hardware and software that is included within the organisation.

Basically, it means that with improved hardware and software, they are much more complicated. The fact that they are much more complicated means that the upgrades over time would take much longer, sometimes being very time consuming.

This longer term proposal could include a plan where the staff can reinstall computer systems or software upgrades a specific time of year, usually when there is less members or when it isn't training time.

If a proposal was to be made when there is no training or very little training, it would be better for the upgrade when compared to the rest of the year. The reason being that the training of the majority of trainees doesn't get disrupted and that all the data and information has a smooth flow without any interference from the upgrades, which can prove to be a disaster.

At least three additional examples of both software and hardware upgrades

When upgrading IT-enabled improvements such as software and hardware, there is hardly a limit to how much you can upgrade to, especially with the release of new products. Also, there is a wide variety of hardware and software that all act differently but some doing the same job as others, meaning that there is plenty room for improvement for an IT-enabled improvement.

One example of an additional hardware upgrade would be to invest for a new fibre optic cabling for the broadband of the organisation. Basically, the fibre optic broadband would increase the bandwidth capacity because optical signals are able to carry much more information than copper wires.

As well as this, signals where fiber optic cabling is used will be much more reliable because they do not suffer from as much electromagnetic interferences as much. Also, fibre optic is much smaller and lighter.

The weight of 1 km of fibre optic cable will be about 6kg but for 1km of coaxial cable, the weight is around 1000kg. Meaning that the cable can be easily moved around and fit into many places where previous wires wouldn't be able to go.

However, fibre optic cabling is very expensive and it is less robust than the traditional copper cabling. If optical fibers are bent, they could lose some of the reflective qualities. It is also easier to damage fiber optic cabling and more expensive and complicated to repair.

Another example of an additional software upgrade is the integration of a FileMaker system. Basically, the FileMaker system would replace the existing system instead of fitting in with it. By replacing the system completely the organisation would be able to monitor trainee progression and add/update classes and training sessions much easier as all of the data would be kept in the same place.

The process of replacing the system is fairly simple as the organisation already has the software and some experience in using it. Once the system has been built, integrating the old system into the new one will be fairly simple as FileMaker allows data to be imported into the database from Excel files.

This will speed up the process of developing the system and allow the managers and staff members to be trained how to use the system before the next training session with the trainees. The new system will either be stored on a FileMaker server, or can be added to the organisation’s central file server where staff can access it.

Finally, the last example of an additional software upgrade is to integrate a VOIP, which means voice over internet protocol, system for the staff, basically giving them the ability to talk to each other without physically being there.

This is a very cheap system as it does not require a regular contract, just the initial installation cost and internet connection. This is particularly cheap considering what the technology has to offer.

The mobility of VOIP will be a huge benefit to Chiltern Global because Chiltern Global is a global company with contacts around the world that it needs to communicate with. THe fact that this technology is not reliant on a phone signal is a huge advantage and means that Chiltern Global will be able to communicate with employees and clients in other areas of the world.

One of the problems with this technology is that it is reliant on power and internet connection. The problem with this is that if there is a power cut or a loss of internet connection, the

technology will be rendered useless. There are also some technical issues regarding the quality of the calls as there is often echoing and static.

Despite some of the improvements that still have to be made and the reliance on an internet connection, this is a worthwhile investment for Chiltern Global because this will allow employees and clients to communicate with very little cost all around the world.

Detailed written accounts of the impacts of each phase of the proposal. i.e. Cost, staff and specific improvements.

With each proposal, there is a type of impact that it can have on each phase, causing the organisation to have to make specific improvements. These improvements can range from cost, staff and other specific improvements such as software and hardware upgrades.

For the cost, I will recommend that they invest in these pieces of hardware and software because the price range is affordable for them and they can integrate it into the systems they already own.

The price of installing these improvements to a good standard would be an estimate of 30k, the reason for this price is because it is made for big businesses that provide tailored service and system for their members.

This looks very pricey, but if the organisation gets the benefits of these upgrades, they could at least break even from the expenses.

For the staff, the impacts is that the organisation has two different departments working together. This would mean that departments will have to work together using business system with less staff. Meaning that the organisation will also have to deal with redundancies, which can have a possible impact to the organisation and the staff.

This is because these improvements in the computer system will then take over some of the work that people will be doing within the organisation, especially with the AI. Those employees will not be needed anymore so the business will have to pay them as they have now been made redundant.

Installing these upgrades will also have an impact on working procedures. Staff will need to turn documents around quickly, also contacting customers will be done by electronically by emails and text messages rather than letters. Contacting customers will become easier and efficient when these upgrades is installed.

For the specific improvements, such as software and hardware upgrades, is that they have to face issues such as maintenance and especially security.

security issues also need to be taken into account. this is because by installing a the hardware and software improvements, the organisation will also need to put in place the correct security methods.

Some security procedures the organisation will have to put in place are secure firewalls to stop hackers trying to get into the network of the organisation and get customer banking details. Also the organisation may need to encrypt files and have a password on them as well.

This way if the organisation was to be hacked, the hacker will have no means of deciphering the encryption, making the stolen information virtually useless.

I believe it will be a good investment for the Jersey Aero Club to install these upgrades into their system. This is because it will take pressure of two departments, the staff training in aviation and the staff teaching in the classroom, in the organisation that will be working together.

It is used to manage the relationship between the organisation and its members. Data stored in the system will also enable the organisation to target marketing campaigns because they can identify customer buying trends.

Basically, The efficiency of the business will be improved once these software and hardware upgrades are installed. Retrieving customer information will be easier and faster, meaning that the organisation could work much more efficiently without worrying about missing information.

(U4:D2) - Fully justify proposals for an IT-enabled improvement.

Should include full justification for your choices and suggestions

I feel that this implementation of the IT-enabled improvement for the Jersey Aero Club organisation would be the right thing to do because it can benefit from the technology which simply was difficult or wasn't possible to do previously.

What I mean by this is that the implementation of the software and hardware that is used by a majority of organisations, can help the organisation to control and improve relationships with data and information with their members.

The software and hardware systems can be installed straight on the organisations facilities. However, most organisations now use internet based programs where all applications and information are stored on the host's server.

The internet based database systems is more reliable as it reduces the IT overhead of the system, increases security of data and reduces the cost hardware and software to companies.

Since the software system can be accessed wirelessly via the internet, this is why I have proposed that we improve it by providing customers services staff with tablets for them to access the company's information system from home instead of them to come into work.

A deep consideration of the Pros & Cons your suggestions

The implementation of the new IT systems may, in theory, be seen as an improvement due to the efficiency of work and income to the business that the IT systems could bring to the company. However, like most things, the implementation of the IT-enabled improvement can have its pros and cons.

The pros that the improvement of software and hardware is that it could improve the flow of information within the organisation. I feel that the improvement of information could be beneficial for an organisation like this because it focuses on training new pilots, and with the information of each trainee is important to have.

The reason for having information is important is due to the fact that the information can hold, the aircraft they own, the staff working, the trainees attending and how well performing and trained the members are. However, this is only the basics of the information that this organisation can have within the database. So improving on this aspect would make the organisation much more efficient.

The cons of the improvement of software and hardware is that it is not guaranteed that this technology would be an improvement for the business. What I mean by this is that, the technology still needs to be perfected, so it is not quite reliable to use yet. Problems such as errors and malfunctions could have a massive impact of the organisation.

The reason being that, if an error or malfunction was to occur, the organisation's information system would have mistakes, therefore creating incorrect information. Which can prove to be extremely problematic for an organisation which has to rely on information in order to do the intended purpose of the organisation itself.

A well justified account in terms of impacts on the organisation, staff, customers and finances

With the improvement for the organisation with hardware and software, there are impacts with the organisation, staff, members and even the finances. However, there are some effects that it will have on the overall organisation due to common problems or problem due to human and technical limitations, but these effects can be overcome and even outweighed by the positives.

For the organisation, we all know that technology is involved in every aspect of the business in today's world. The level that Total Control is now, compares to other businesses, we are far behind.

This is why I think it will be a good idea to improve the hardware and software systems by connecting it the mini tablets. Even though I know that it will be a big risk and a huge step to take from where we are now, but we have to do because if we don't the organisation might not exist anymore in the next 2 years, due to it's inefficient method of gathering information.

For the staff, they will be under a lot of pressure as they have to learn new skills and use the unfamiliar technology immediately, which can be hard for them. They might make some mistakes which can make them lose confidence in themselves. Also, it's possible that it might not recover the cost of the project for example if staffs stop working.

For the members, The company might also think of installing a method of augmented reality in the classes which will be connected electronically to the database system which will display the information as the orders are being recorded and modified in order to increase the speed of work.

It also be good if the aircraft supervisors could have access to the computer from the warehouse to update the information about tasks in the database system in order for other departments to be up to date with the progress of the work. Basically, the organisation should think about having their own advanced database system in future if they want to make more profit.

With finance, the problem that the organisation might face can be losing staff because they are not happy to do the training and decide to leave simply for their personal reasons or any other reasons which can have many possibilities for.

If this happens the company will not only lose the employee, but also the value of that employee has, which is the experience. This can cost the company lots of money to replace them as the new employee must be trained in order to have the experience.

Consideration legal/regulatory frameworks

With the installation of this new software and hardware, there are some consideration of legal and regulatory frameworks which the organisation has to negotiate and respond to. Whether that is the handling of information or the health and safety of staff and trainees.

When handling information, the organisation will have to make sure that it fits within the legal framework. The reason for this is because the handling of illegal or having information which was not stated in a contract would be punished by the law, causing the company to be shut down, of if the company is lucky, get a large fine.

With the health and safety of personnel when using the hardware, the organisation legally needs to check whether the hardware is safe to use. If it isn't, and a number of staff or another

trainee gets injured from that cause. The organisation would have to pay for health insurance or also face the law and again get the risk of being shut down.

Overall, if the organisation does not consider the legal and regulatory frameworks, it would give the organisation a bad image to the public due to its illegal acts which would be reported. Which means that the organisation will have a very miniscule chance in being successful, which means that they will eventually have to shut down the organisation.

Security issues to negotiate and respond to

With the installation of this new software and hardware, there are some security issues which the organisation has to negotiate and respond to. Whether that is the protection against hackers or other forms of malware which could significantly harm the network of the organisation.

With protection against hackers, the organisation will have to make sure that it is safe from unknown personnel planning to intercept the data. The reason for this is because the loss of data within a database due to it being stolen can be catastrophic for the business, the reason being that the organisation will have to remake that data and also be worried with the fact that a potential competitor has stole valuable information.

With the protection against malware and other harmful computer viruses, the organisation will need to check whether the software and its data is protected from a malware attack. If it isn't, there will be a risk of a virus intercepting the computer systems and corrupting the files within the computer, either rendering it useless or removing all the files.

Overall, if the organisation does not consider the protection against malware and other forms of harmful computer software, the organisation would have a high chance of losing all its data. Which means that the organisation will have to reattain that data from the start, which would annoy the members of staff and trainees, risking them to leave the organisation and leaving it vulnerable to bankruptcy.

Specific efficiency gains the organisation will experience

With the installation of this new software and hardware, there is no doubt that the organisation will experience a form of an efficiency gain. Which is mostly affected within the flow of data and the training of the members.

When managing the flow of data, the organisation would have increased in efficiency due to the new software and hardware installations. The reason for this is that the new hardware and software such as mobile devices, advanced database software, better computer systems and faster internet speeds, means that the flow of data would be significantly faster and less likely to have a malfunction, which means it's more efficient.

With the training of the members of the JAC, the improvement of the software and hardware can prove to be more efficient with the staff teaching the beginner pilots. With the hardware such as AR, advanced surveillance and even a simple CPS, can improve the teaching of the trainees. Yes, this may benefit the flow of information, but with that information the staff can use it to train the beginner pilots that are in need, instead of the ones that are performing very well.

Overall, the implementation of the hardware and software can prove to be very efficient for the teaching and flow of information within the organisation. Which means that the organisation will be much faster and of higher quality when compared to before, which would increase the reputation of the organisation, which can convince a lot more members to join and make the organisation profit more.

Task 2

(U3:P6) - Select information to support a business decision-making process.

Provide details of at least two software tools required for gathering information from the business

Undoubtedly, gathering information is very important for an organisation due to the fact that information can be used by the business to find out about employees and customers, but to gather information the business will need tools to carry out the job.

One of these tools can be a database software, since it is the intended purpose of a database to gather and contain information for the organisation. So, this software wouldn't be a problem to use to gather information and even store it for future use.

Another form of these tools can be the applications within the mobile devices, due to the mobile devices being able to store many types of information and applications like a PC could, it could have applications in which it can store data which it later can be stored within a professional database software.

Define the information the business may be required to gather by law

Gathering information is a very sensitive topic, both for the business and the personnel within the organisation, especially the personnel in control. The reason being is that the law specifically prevents a company or person handling certain types of information, and that only the stated information could be held.

This means the information has to fit within the requirements of the law and only to contain information which the organisation can only use for the benefit of the organisation and the members of said organisation.

With the data protection law of the EU, saying that the data held should: data subjects should be given notice when their data is being collected should fit within these standards; data should only be used for the purpose stated and not for any other purposes; data should not be disclosed without the data subject's consent; collected data should be kept secure from any potential abuses; data subjects should be informed as to who is collecting their data; data subjects should be allowed to access their data and make corrections to any inaccurate data and data subjects should have a method available to them to hold data collectors accountable for not following the above principles

This means that the type of information which the organisation can legally have ranges from the names of the members, their age, their address and phone number for contacting and how they are performing within the organisation.

However, this data wouldn't still be legal to have without the authorisation of the member his/her self, meaning that the organisation would have to make a contract saying that they can own any of the stated information which is shown on the contract. Which would make it legal to contain the information the organisation promised to only have of that person.

Establish at least three separate sources of information in the business

When gathering information within the business itself, there are many sources in which the business can gather information. For instance, the separate sources that the business could collect information from could either be an employee questionnaire issued by the manager, daily employee feedback to a manager or a customer report from surveillance.

With a questionnaire, the organisation is able to ask specific questions which they feel would be important to answer and have benefits to the future of the organisation. The reason being that this type of information, if collected correctly, can be used to improve any problems or errors within the organisation which can be resolved.

With these questions, they could be answered either by a member of staff or a trainee member of the JAC, in which the answers from each of these members could be used as information for future benefits for the organisation.

With employee feedback to a manager, the organisation is able to receive feedback from the employees on how the new members are doing and feedback about the working conditions, which can also have benefits to the future of the organisation.

With this feedback, the organisation could find out any problems with the staff, working conditions or the members and address that issue as soon as possible, to make the organisation a much better place of work and more reputable for the staff and members of the JAC.

With member report from surveillance, the organisation is able to track the members within the business and how they are performing. This is type of information is vital because it is what the organisation is made for, which is to train beginner pilots and for them to get their license.

So, with these reports, the organisation could use the information gathered to grade and rank the pilots appropriate to their level of skill and knowledge. Which can make the organisation complete its intended purpose much more efficiently and it also looks like a very reputable business in the eyes of the staff and members.

Establish at least two separate sources of information from outside the business

When gathering information outside a business, there also are many sources in which the business can gather information. For instance, the separate sources that the business could collect information from could either be through feedback in social media sites or an online survey.

With feedback from social media, the organisation is able to receive feedback on how the new members or members of staff are doing, about the working conditions and even who ever wants to be recruited, which can again have benefits to the future of the organisation.

With this feedback, the organisation could again find out any problems with the staff, working conditions, the members and the people who want to join the JAC. Which they address the issues and other important acts as soon as possible, to make the organisation a much better place of work, more reputable for the staff and members and to grow the organisation.

With an online survey, it is exactly like the questionnaire, but it's to people outside the organisation. With this, the organisation is able to ask specific questions which they feel would be important to answer and have benefits to the future of the organisation. The reason being that this type of information, if collected correctly, can be used to improve any problems, errors within the organisation which can be resolved or even ask about whether the user would want to be recruited.

With these questions, they could be answered either by a member of staff outside of work, a trainee member of the JAC or even a user who just found out about the organisation. Which the answers from each of these members could be used as information for future benefits for the organisation, such as improvements and growth of the organisation.

Explain any potential constraints or problems the business my need to overcome

With the installation of newer technology to gather information for the business, there is a possible chance that there might be problems which come with the gathering the information or with the software itself.

One problem which the organisation may face is the handling or gathering of incorrect or lack of sufficient data. Which can be a significant problem if the organisation hold insufficient data and doesn't overcome the issue, making the organisation very hard to maintain and give a very bad reputation for the personnel which is a part of the organisation.

It's not always guaranteed that data will always be correct or useful, which means that the business would have to make some adjustments or measures to make sure that the information that is being collected is correct and sufficient enough to be useful.

Another constraint the organisation may face is the legal constraints, especially with the data that is being handled. When handling information, there isn't only technical problems which the organisation can face, there are legal constraints too.

With legal issues such as handling illegal data, property damage, product damage and even the injury of personnel. All these constraints are important due to health and safety, but the one which is critically important in a legal aspect is the handling of illegal data.

If the organisation were to handle illegal data, the law would have to take action on the organisation if caught, and it would have to face the consequences of handling illegal information without it being removed. Which would cause the organisation to be shut down or to have an extremely bad reputation for the public, limiting the possibilities of the business making a profit.

(U3:D2) - Justify the information selected to support a business decision-making process.

Detailed reasons for selecting the Software tools for information gathering

There are many reasons in which selecting the correct and efficient software tools for information gathering could be impactful on a business and its income. These reasons can range from software efficiency, performance, wide amount of information storage and other benefits which a unique piece of software could improve information storage and management.

With software efficiency, the advanced database software has a massive performance improvement when compared to other forms of database, such as MS excel, especially with a high end PC.

The reason for this is that the new database software, it has more advanced features and is much more optimised because it is of premium quality when compared to free applications such as MS excel, this means that the flow and the storage of data would be significantly faster and less likely to crash, making it much more efficient.

With the performance of the software, the improvement of the software can prove to be a lot faster due to its refined, premium quality build when compared to free applications. With the vast amount of tools and optimisation to store data, it can improve the flow of information through a computer system.

With the wide amount of information storage, the improvement of the software also benefits the wide amount of storage within the database without it having the risk of crashing. Which means that there is a less chance of all the data that is recorded to be lost and have to be regained, which could be problematic for the organisation.

Overall, the implementation of the advanced database software can prove to be very efficient for the flow and storage of information within the organisation. Which means that the organisation will be much faster and of higher quality when compared to before due to the importance of information, which would increase the reputation of the organisation.

Justification for the internal and external sources of information you have picked

With the internal and external sources of information that I have picked, I feel that there is good reasoning on why I have picked them. These reasons can range from useful information, reducing the lack of information, able to keep information without getting it mistaken and other benefits which the company could improve from.

With useful information, the methods of getting the information that I have stated I believe to be a good way of collecting information, but no where near the best way of doing so.

The reason for this is that there isn't a guaranteed way to always get useful information, but it does mean that there is less of a risk of getting information that is not needed, this means that the information can easily be stored without worrying about it being not so useful, with those that aren’t useful, they could just be ignored.

With reducing the lack of information, the method in which I gave I believe that it would reduce the risk of getting the lack of information, the reason being is because there are some required field within the questionnaires and queries. Meaning that it would have to be filled in or it cannot be sent, which means that almost any form of feedback the organisation receives would contain information, but it’s not guaranteed if it is useful or not.

With the ability to keep information without getting it wrong, the method in which I gave I believe that it would reduce the risk of getting the information that is given wrong, the reason being is because the sources of information which can come from staff, members of the JAC or other people who want to be recruited, would most likely take the surveys and questionnaires seriously. Meaning that they would fill in the information sheets without lying, which also means that almost any form of feedback the organisation receives would contain useful information.

Overall, the methods of receiving information from within and outside the organisation can prove to be worthwhile, but not perfect. However, it does mean that the organisation will be more likely to get useful and serious information from the feedback of users, which the organisation could benefit from.

Reasons for the need to overcome constraints you have identified

With the constraints that I have identified, I feel that overcoming them would benefit the company significantly. These reasons can range from incorrect data, lack of useful data, legal constraints and other constraints which could affect the organisation massively.

With incorrect data, the organisation will have to make sure that it doesn't contain incorrect data, and by doing this they need to have to have a form of validation to make sure it is correct. The reason for this is because having incorrect of data within a database can be catastrophic for the business, the reason being that the organisation will use that data and not know it is incorrect, which can cause significant problems such as incorrect software, incorrect members for incorrect courses or even the incorrect staff to teach in aviation.

With the lack of useful data, the organisation will have to overcome this problem because having no information is almost as bad as having no information, due to the organisation not knowing what to do. Without information, there will be a risk that the business would have to cancel the class and members within the organisation, due to them having no reason on being there because there is no information to obtain from them.

With the legal constraints and other constraints which could affect the organisation, the organisation will need to consider the legal constraints and other constraints in which it could affect the organisation's reputation and legal stance. If the organisation does not keep this in mind, there will be a risk of an illegal act within the organisation which no one will notice, and once the law find out, the organisation would have to face the consequences and will have a bad reputation for the public.

Overall, if the organisation does not consider the constraints that I have identified, the organisation would have a high chance of losing its members, staff and even profits. Which means that the organisation will have to shut down, which would be a major inconvenience to those who wanted to use the organisation well, due to the business going bankrupt or being shut down by the law.